

The Language of Medicine

13th edition

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Chapter 14

Lymphatic and Immune System

Chapter Goals (Slide 1 of 2)

- Identify the structures and analyze terms related to the lymphatic and immune systems.
- Learn basic terminology, combining forms, and other word parts related to these systems.
- Recognize terms describing pathologic conditions.

Chapter Goals (Slide 2 of 2)

- Identify laboratory tests, clinical procedures, and abbreviations.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 14

Lesson 14.1

Introduction (Slide 1 of 2)

Lymph: clear watery fluid that surrounds body cells and flows in a system of lymph vessels throughout the body

- Originates in blood
- Rich in lymphocytes and monocytes
- Flows in lymph capillaries and vessels
- Flows through lymph nodes and lymphatic organs (liver, spleen, thymus, and nodes)
- Absorbs lipids in the intestine

Introduction (Slide 2 of 2)

Lymphatic System Functions

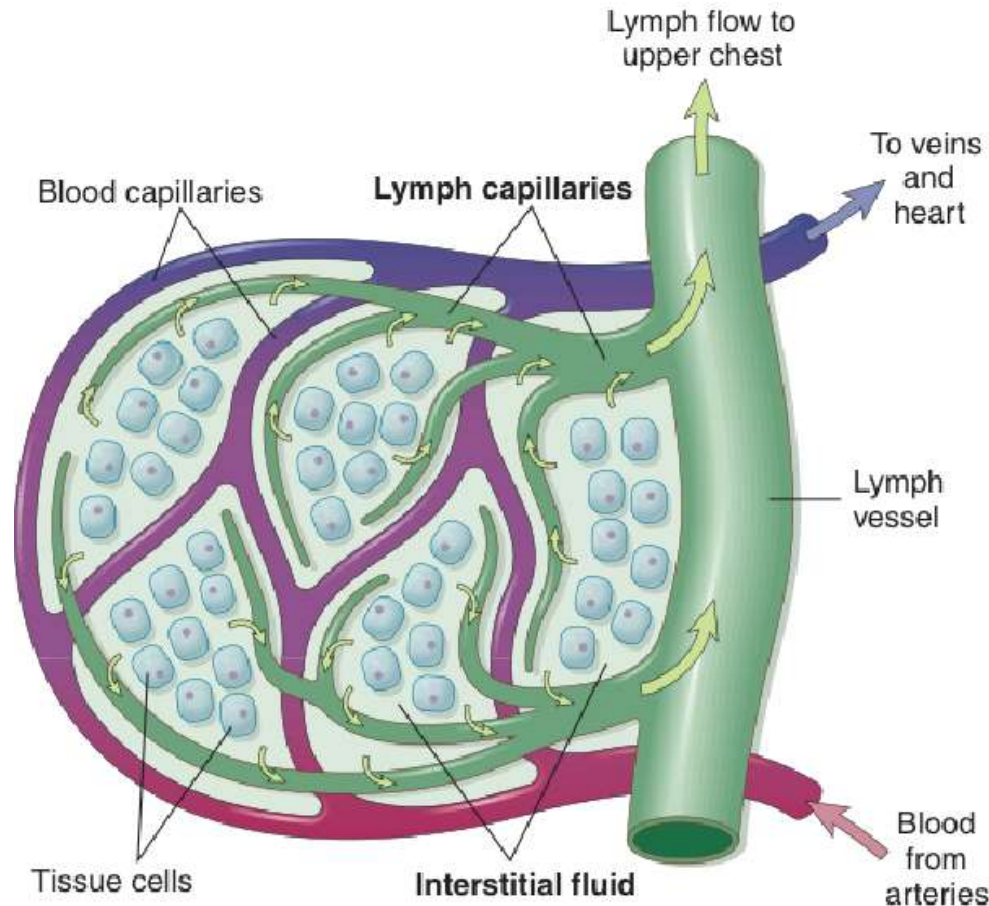
1. Transports proteins and fluid that have leaked, back to the bloodstream
2. Lymphatic vessels absorb lipids from intestines and transport them to the bloodstream
3. Lymphocytes and monocytes protect the body by mounting a cellular attack on foreign cells and organisms.

Interstitial Fluid and Lymph Capillaries

(Slide 1 of 2)

- Structures of interstitial fluid and lymph capillaries include:
 - **Blood capillaries**
 - **Lymph capillaries**
 - **Lymph vessel**
 - **Interstitial fluid**
 - **Tissue cells**

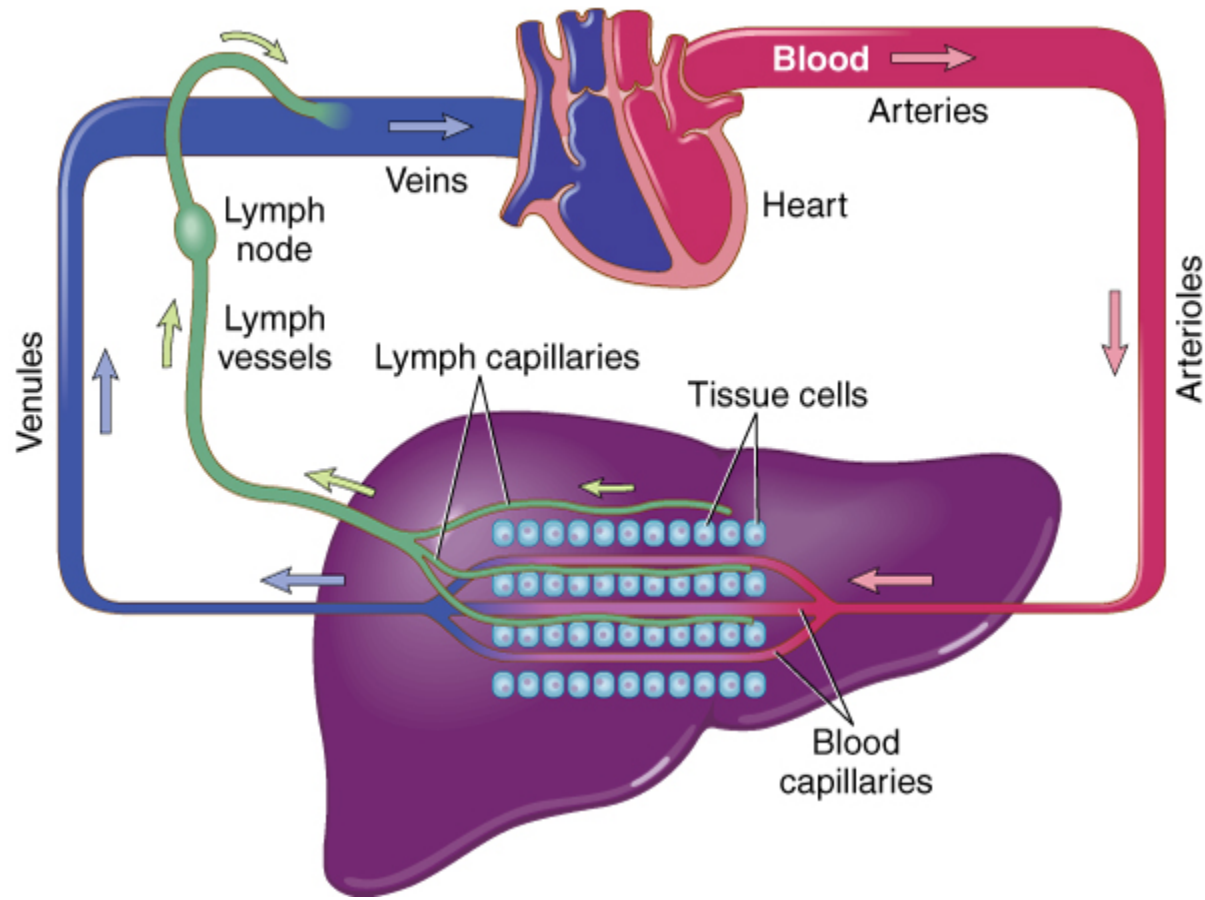
Interstitial Fluid and Lymph Capillaries (Slide 2 of 2)



Blood and Lymph (Slide 1 of 2)

- Lymph originates from the blood and filters out of tiny blood capillaries into the spaces between cells.
- Fluid that surrounds body cells is called interstitial fluid, which passes continuously into thin-walled vessels called lymph capillaries that course through tissue spaces.
- Lymph passes through larger lymphatic vessels through clusters of lymph tissues, finally reaching large lymphatic vessels in the upper chest.
- Lymph enters these large lymphatic vessels, which then empty into the bloodstream.

Blood and Lymph (Slide 2 of 2)



Lymphatic System Anatomy

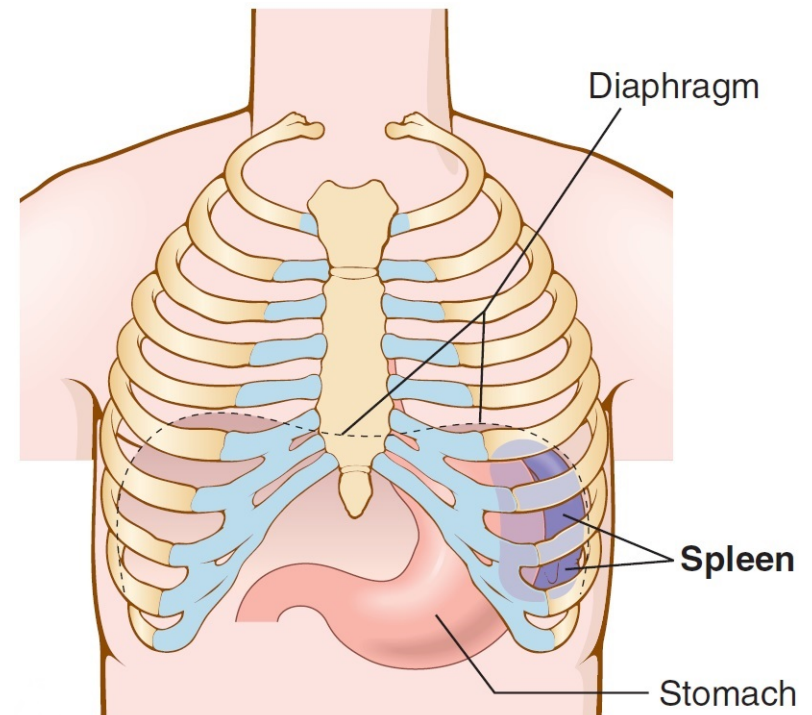
- Lymph capillaries
- Lymph vessels
- Lymph nodes
 - Cervical (neck)
 - Axillary (armpit)
 - Mediastinal (chest)
 - Mesenteric (intestinal)
 - Paraaortic (lumbar)
 - Inguinal (groin)

Lymph Nodes

- **Macrophages** phagocytose foreign substances
- **B lymphocytes (B cells)** produce antibodies; mature in the bone marrow
- **T lymphocytes (T cells)** attack bacteria and foreign cells; originate in the thymus gland

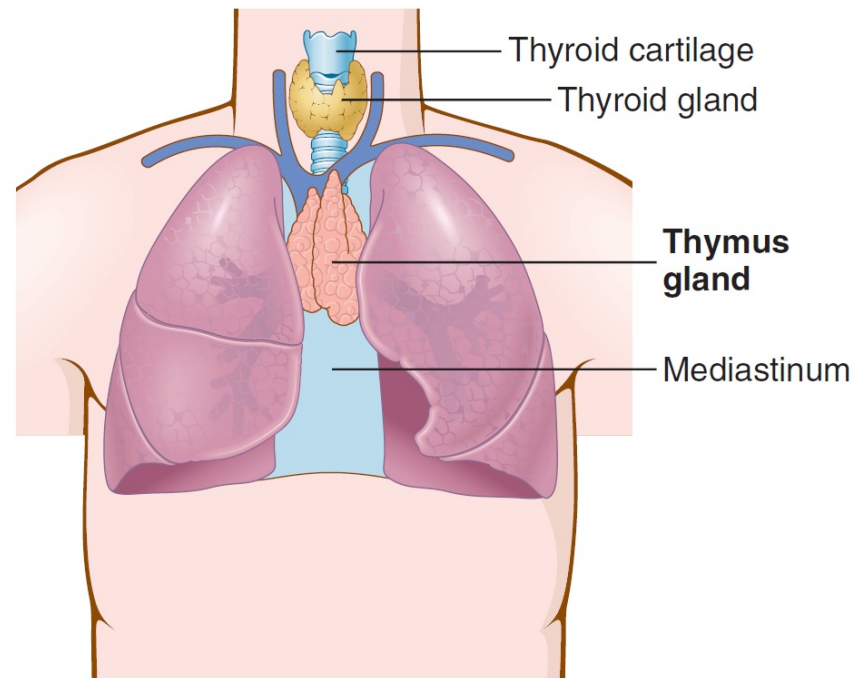
Spleen

- Destroys old red blood cells
- Filters microorganisms and other foreign material out of the blood
- Activates lymphocytes during blood filtration (Activated B lymphocytes (B cells) produce antibodies; Activated T lymphocytes (T cells) attack foreign materials)
- Stores blood, especially erythrocytes (RBCs) and platelets



Thymus Gland

- Provides immunity in fetal life and in early years of growth
- Makes cells immunocompetent in early life
- Early removal from an animal impairs its ability to make antibodies and produce cells to fight antigens



Immune System

- Leukocytes: neutrophils, monocytes, and macrophages
- Lymphoid organs: lymph nodes, spleen, thymus gland, tonsils, and adenoids; produce lymphocytes and antibodies
- **Immunity:** the body's ability to resist foreign organisms and toxins that damage tissues and organs; can be natural or adaptive

Natural Immunity

Natural immunity: resistance present at birth;
not dependent on prior exposure to an
antigen

- Phagocytosis
- Macrophages
- Natural killer cells

Adaptive Immunity (slide 1 of 4)

Adaptive immunity: body's ability to recognize and remember specific antigens in an immune response:

- Lymphocytes: T and B cells
- Antibodies produced after exposure to antigen
- Memory: second exposure
- Vaccination
- Immunoglobulins
- Maternal antibodies

Adaptive Immunity (Slide 2 of 4)

B lymphocytes (B cells) are responsible for humoral immunity.

- Originate in bone marrow from stem cells
- B cells produce antibodies after exposure to specific antigens (viruses and bacteria)
- B cell matures into plasma cell that produces antibodies called immunoglobulins to block the effects of antigens

Adaptive Immunity (Slide 3 of 4)

T lymphocytes (T cells) are responsible for cell-mediated immunity; does not involve antibodies

- Cytotoxic T cells (CD8+ T cells) attach to and directly kill antigens.
- Other T cells secrete cytokines (interferons and interleukins) that help cells respond to antigens.
- Helper T cells (CD4+ T cells) promote antibody synthesis by B cells and stimulate other T cells.
- Suppressor T cells (regulatory T cell or Tregs) inhibit B and T cells and prevent them from attacking the body's own good cells.

Adaptive Immunity (Slide 4 of 4)

Dendritic cells

- Dendritic cells are a macrophage derived from monocytes.
- They recognize and digest foreign antigens.
- They present antigens on the surface to stimulate B and T cells.
- They transfer immunity by exposing dendritic cells in culture; then transfuse them and get them to stimulate T and B cells.

Immunotherapy Examples

- **Monoclonal antibodies (MoAb):** multiple copies of cells or genes that kill tumor cells; passive immunotherapy
- **Vaccines:** contain antigens from the patient's tumor cells; stimulate the patient's own T cells to recognize and kill the cancerous cells; active immunotherapy
- **Transfer of immune cells:** In bone marrow transplantation, T lymphocytes from a donor can replace a patient's immune system with new cells that recognize tumor cells as foreign and kill them.

Vocabulary (Slide 1 of 30)

Term
adaptive immunity
adenoids
antibody

Vocabulary (Slide 2 of 30)

Term	Meaning/Definition
adaptive immunity	Ability to recognize and remember specific antigens and mount an attack on them
adenoids	Mass of lymphatic tissue in the nasopharynx
antibody	Protein produced by B cells to destroy antigens

Vocabulary (Slide 3 of 30)

Term
antigen
axillary nodes
B cell (B lymphocyte)

Vocabulary (Slide 4 of 30)

Term	Meaning/Definition
antigen	Substance that the body recognizes as foreign; evokes an immune response
axillary nodes	Lymph nodes in the armpit
B cell (B lymphocyte)	Lymphocyte that matures into a plasma cell to secrete antibodies

Vocabulary (Slide 5 of 30)

Term
cell-mediated immunity
cervical nodes
complement system

Vocabulary (Slide 6 of 30)

Term	Meaning/Definition
cell-mediated immunity	T cells that respond to antigens and destroy them
cervical nodes	Lymph nodes in the neck region
complement system	Proteins in the blood that help antibodies and kill their target

Vocabulary (Slide 7 of 30)

Term
cytokines
cytotoxic T cell (CD8+ T cell)
dendritic cell

Vocabulary (Slide 8 of 30)

Term	Meaning/Definition
cytokines	Proteins secreted by cytotoxic T cells to aid in antigen destruction
cytotoxic T cell (CD8+ T cell)	Lymphocyte that directly kills antigens
dendritic cell	Antigen-presenting cell; shows T and B cells what to attack

Vocabulary (Slide 9 of 30)

Term
helper T cell (CD4+ T cell)
humoral immunity
immunity

Vocabulary (Slide 10 of 30)

Term	Meaning/Definition
helper T cell (CD4+ T cell)	Lymphocyte that aids B cells and stimulates T cells
humoral immunity	B cells produce antibodies after exposure to specific antigens; type of adaptive immunity
immunity	Body's ability to resist foreign organisms and toxins that damage tissues and organs

Vocabulary (Slide 11 of 30)

Term
immunoglobulins
immunotherapy
inguinal nodes

Vocabulary (Slide 12 of 30)

Term	Meaning/Definition
immunoglobulins	Antibodies secreted by plasma cells in response to the presence of an antigen
immunotherapy	Use of immune cells, antibodies, or vaccines to treat or prevent disease
inguinal nodes	Lymph nodes in the groin region

Vocabulary (Slide 13 of 30)

Term
interferons
interleukins
interstitial fluid

Vocabulary (Slide 14 of 30)

Term	Meaning/Definition
interferons	Proteins secreted by T cells and other cells to aid and regulate the immune response
interleukins	Proteins that stimulate the growth of B and T lymphocytes
interstitial fluid	Fluid in the spaces between cells; becomes lymph in lymph capillaries

Vocabulary (Slide 15 of 30)

Term
lymph
lymph capillaries
lymphoid organs

Vocabulary (Slide 16 of 30)

Term	Meaning/Definition
lymph	Thin, watery fluid within lymphatic vessels and collected from tissues throughout the body
lymph capillaries	Tiniest lymphatic vessels
lymphoid organs	Lymph nodes, spleen, and thymus gland

Vocabulary (Slide 17 of 30)

Term
lymph node
lymph vessel
macrophage

Vocabulary (Slide 18 of 30)

Term	Meaning/Definition
lymph node	Collection of stationary solid lymphatic tissue along lymph vessels
lymph vessel	Carrier of lymph throughout the body
macrophage	Large phagocyte found in lymph nodes and other tissues of the body

Vocabulary (Slide 19 of 30)

Term
mediastinal nodes
mesenteric nodes
monoclonal antibody

Vocabulary (Slide 20 of 30)

Term	Meaning/Definition
mediastinal nodes	Lymph nodes in the area between the lungs in the thoracic cavity
mesenteric nodes	Lymph nodes in the mesentery
monoclonal antibody	An antibody produced in a laboratory to attack antigens and to destroy cells

Vocabulary (Slide 21 of 30)

Term
natural immunity
paraaortic nodes
plasma cell

Vocabulary (Slide 22 of 30)

Term	Meaning/Definition
natural immunity	Protection that an individual is born with to fight infection
paraaortic nodes	Lymph nodes near the aorta in the lumbar area of the body
plasma cell	Lymphocyte that secretes antibodies; matures from B lymphocytes

Vocabulary (Slide 23 of 30)

Term
Right lymphatic duct
Spleen

Vocabulary (Slide 24 of 30)

Term	Meaning/Definition
Right lymphatic duct	Vessel in the chest that drains lymph from the upper right part of body; empties into large neck vein
Spleen	Organ in the left upper quadrant of the abdomen that destroys worn-out RBCs, activates lymphocytes, and stores blood

Vocabulary (Slide 25 of 30)

Term
suppressor T cell (Treg)
T cell (T lymphocyte)
tolerance

Vocabulary (Slide 26 of 30)

Term	Meaning/Definition
suppressor T cell (Treg)	Lymphocyte that inhibits the activity of B and T cells
T cell (T lymphocyte)	Lymphocyte that acts directly on antigens to destroy them or produces chemicals toxic to antigens
tolerance	Ability of T lymphocytes to recognize and accept the body's own antigens as "self" or friendly

Vocabulary (Slide 27 of 30)

Term
thoracic duct
thymus gland
tonsils

Vocabulary (Slide 28 of 30)

Term	Meaning/Definition
thoracic duct	Drains lymph from the lower and left side of body; empties into large veins in neck
thymus gland	Lymphoid organ in the mediastinum that conditions T cells to react to foreign cells
tonsils	Masses of lymphatic tissue in back of the oropharynx

Vocabulary (Slide 29 of 30)

Term
toxin
vaccination
vaccine

Vocabulary (Slide 30 of 30)

Term	Meaning/Definition
toxin	Poison; a protein produced by certain bacteria, animals, or plants
vaccination	Exposure of an individual to an antigen that provokes an immune response that will destroy any cell with the antigen and will protect against infection
vaccine	Weakened or killed microorganisms, toxins, or other proteins given to induce immunity to infection or disease

QUICK QUIZ (Slide 1 of 5)

- 1. Which lymphocyte originates in the bone marrow and transforms into a plasma cell to secrete antibodies?**
 - A. Dendritic cell
 - B. Helper T cell
 - C. B cell
 - D. Cytotoxic T cell

QUICK QUIZ (Slide 2 of 5)

2. The formation of antibodies and lymphocytes after exposure to an antigen is:

- A. cell-mediated immunity
- B. humoral immunity
- C. natural immunity

Combining Forms, Prefixes, and Terminology (Slide 1 of 2)

Combining Form	Meaning
immun/o	protection
lymph/o	lymph
lymphaden/o	lymph node
splen/o	spleen
thym/o	thymus gland
tox/o	poison

Combining Forms, Prefixes, and Terminology (Slide 2 of 2)

Prefix	Meaning
ana-	again, anew
inter-	between

Chapter 14

Lesson 14.2

Disorders of the Lymphatic and Immune Systems

Immunodeficiency

- Acquired immunodeficiency syndrome (AIDS)
 - Destroys T helper cells (CD4+ cells)
 - Opportunistic infections
 - Malignancies
(Kaposi sarcoma, wasting syndrome)

Opportunistic Infections with AIDS

- Candidiasis
- Cryptococcus (Crypto)
- Cryptosporidiosis
- Cytomegalovirus (CMV) infection
- Herpes simplex
- Histoplasmosis (Histo)
- *Mycobacterium avium-intracellulare (MAI) infection*
- Pneumocystis pneumonia (PCP)
- Toxoplasmosis (Toxo)
- Tuberculosis (TB)

Drugs to Treat AIDS

- **Reverse transcriptase inhibitors (RTI)** inhibit viral enzyme called reverse transcriptase.
- **Protease inhibitors** inhibit viral proteolytic enzyme.
- Combination of RTI and protease inhibitors is **HAART** (highly active anti-retroviral therapy).

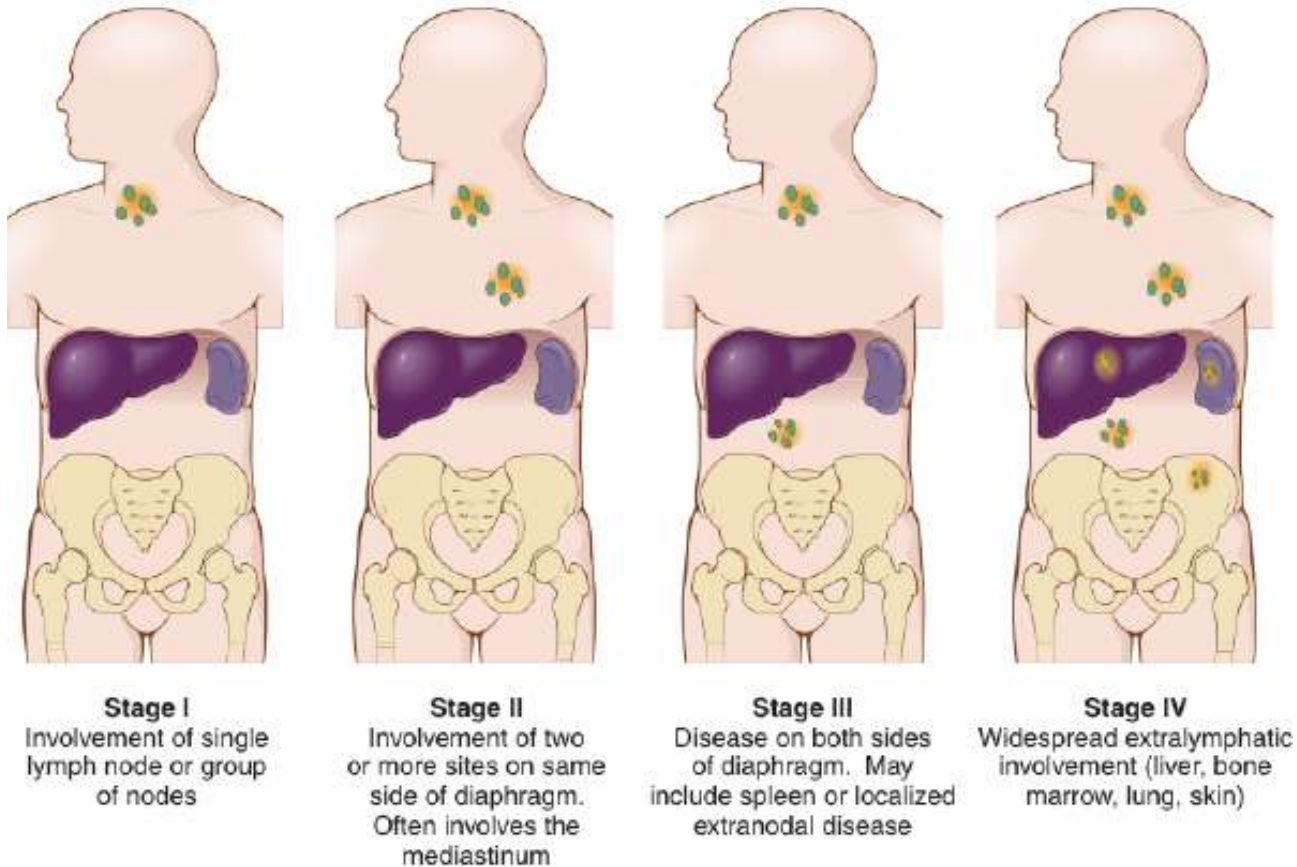
Hypersensitivity

- **Allergy:** abnormal sensitivity acquired by exposure to antigen
 - Ranges from
 - Allergic rhinitis or hay fever to anaphylaxis
 - Other allergies:
 - Atopic dermatitis
 - Asthma
 - Urticaria (hives)

Tumors

- **Lymphoma:** malignant tumor of lymph nodes and lymph tissue
 - Hodgkin lymphoma (Reed-Sternberg cell)
 - Non-Hodgkin lymphoma (follicular lymphoma, large cell lymphoma)
- **Multiple myeloma:** malignant tumor of bone marrow cells
- **Thymoma:** tumor of the thymus gland
 - Associated with myasthenia gravis
 - Patients may have other autoimmune disorders such as lupus erythematosus and rheumatoid arthritis

Staging of Hodgkin Lymphoma



QUICK QUIZ (Slide 3 of 5)

3. What is a malignant condition associated with AIDS?

- A. Anaphylaxis
- B. Non-Hodgkin lymphoma
- C. Kaposi sarcoma
- D. Hodgkin disease

Laboratory Tests

- **CD4+ cell count:** measures the number of CD4+ T cells (helper T cells) in the bloodstream with patients with AIDS
- **ELISA test:** detects anti-HIV antibodies; Western blot given as follow-up
- **Immunoelectrophoresis:** separates immunoglobulins (IgG, IgM, IgE, IgA, IgD)
- **Viral load test:** measures the amount of AIDS virus (HIV) in the bloodstream.

Clinical Procedure

- **Computed tomography (CT) scan**

X-ray imaging produces cross-sectional and other views of anatomic structures. These x-ray views show abnormalities of lymphoid organs, such as lymph nodes, spleen, and thymus gland.

Abbreviations (Slide 1 of 3)

Abbreviation	Meaning
AIDS	Acquired immunodeficiency syndrome
CD4+ cell	Helper T cell
CD8+ cell	Cytotoxic T cell
CMV	Cytomegalovirus
Crypto	Cryptococcus
ELISA	Enzyme-linked immunosorbent assay
G-CSF	Granulocyte colony-stimulating factor
GM-CSF	Granulocyte-macrophage colony-stimulating factor
HAART	Highly active antiretroviral therapy

Abbreviations (Slide 2 of 3)

Abbreviation	Meaning
Histo	Histoplasmosis
HIV	Human immunodeficiency virus
HSV	Herpes simplex virus
IgA, IgD, IgE, IgG, IgM	Immunoglobulins
IL38 to IL5	Interleukins
KS	Kaposi sarcoma
MAI	Mycobacterium avium-intracellulare (MAI) complex
MoAb	Monoclonal antibody

Abbreviations (Slide 3 of 3)

Abbreviation	Meaning
NHL	Non-Hodgkin lymphoma
PCP	Pneumocystis pneumonia
PI	Protease inhibitor
RTI	Reverse transcriptase inhibitor
SCID	Severe combined immunodeficiency disease
Treg	Regulatory T cell (suppressor T cell)
Toxo	toxoplasmosis

Review Sheet: Combining Forms (Slide 1 of 4)

Combining Form
axill/o
cervic/o
immun/o
inguin/o
lymph/o
lymphaden/o

Review Sheet: Combining Forms (Slide 2 of 4)

Combining Form	Meaning
axill/o	armpit
cervic/o	neck; cervix (neck of uterus)
immun/o	immune; protection; safe
inguin/o	groin
lymph/o	lymph
lymphaden/o	lymph gland (node)

Review Sheet: Combining Forms (Slide 3 of 4)

Combining Form
splen/o
thym/o
tox/o

Review Sheet: Combining Forms (Slide 4 of 4)

Combining Form	Meaning
splen/o	spleen
thym/o	thymus gland
tox/o	poison

Review Sheet: Suffixes

(Slide 1 of 4)

Suffix
-cytosis
-edema
-globulin
-megaly
-oid
-pathy

Review Sheet: Suffixes

(Slide 2 of 4)

Suffix	Meaning
-cytosis	condition of cells; slight increase in numbers
-edema	swelling
-globulin	protein
-megaly	enlargement
-oid	resembling
-pathy	disease; emotion

Review Sheet: Suffixes

(Slide 3 of 4)

Suffix
-penia
-phylaxis
-poiesis
-stitial
-suppression

Review Sheet: Suffixes

(Slide 4 of 4)

Suffix	Meaning
-penia	deficiency
-phylaxis	protection
-poiesis	formation
-stital	to set; pertaining to standing or positioned
-suppression	to stop

Review Sheet: Prefixes

(Slide 1 of 2)

Prefix
ana-
auto-
hyper-
inter-
retro-

Review Sheet: Prefixes

(Slide 2 of 2)

Prefix	Meaning
ana-	up; apart; backward; again; anew
auto-	self; own
hyper-	above; excessive
inter-	between
retro-	behind; back; backward

QUICK QUIZ (Slide 4 of 5)

4. What is the name of the fluid that lies between cells throughout the body?

- A. Atopy
- B. Lymphadema
- C. Interstitial
- D. Lymphopoiesis

QUICK QUIZ (Slide 5 of 5)

- 5. Malignant tumor of the lymph nodes and spleen marked by the presence of Reed-Sternberg cells in lymph nodes:**
- A. HIV
 - B. Kaposi sarcoma
 - C. Hodgkin lymphoma
 - D. Hodgkin disease